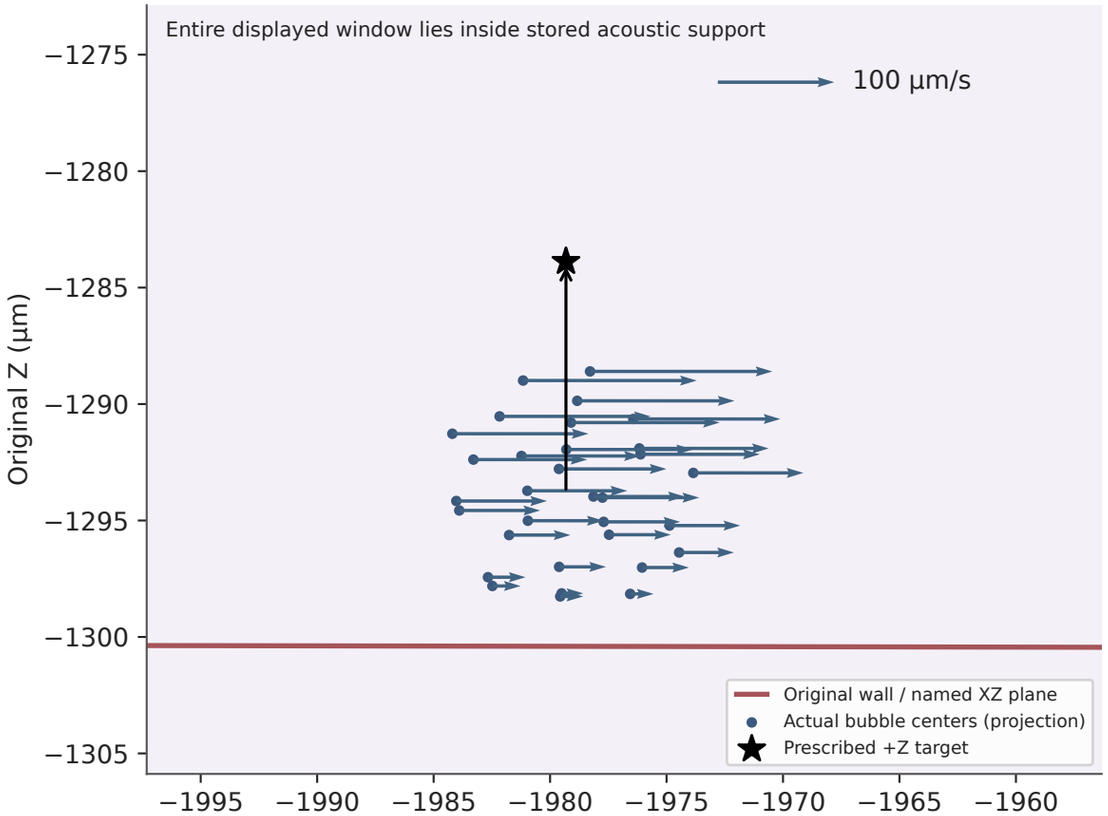
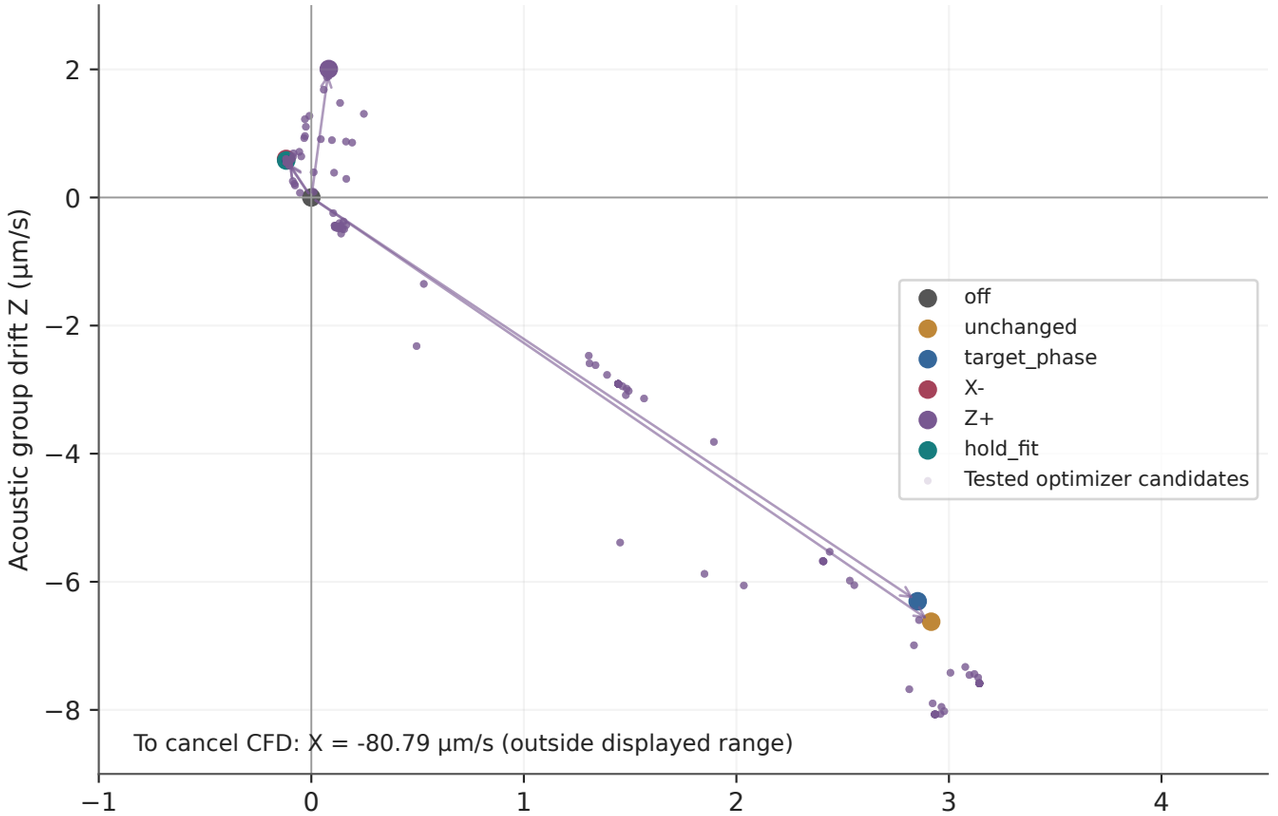
Original Y-vessel: fixed-target reachability diagnosis

(a) Original Y local region; wall plane Y = 0.886 µm

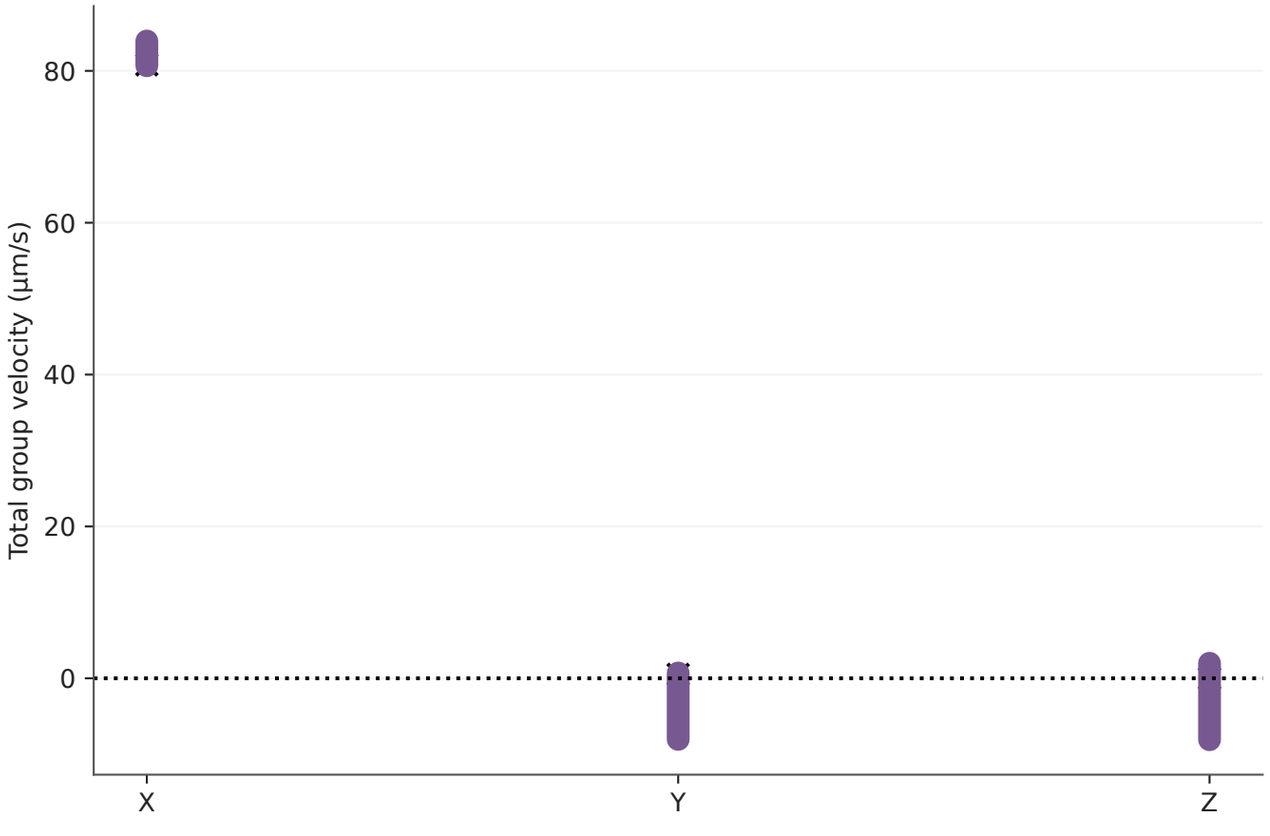


(b) RPY acoustic authority: XZ projection



Original X (µm)

(c) Observed candidate ranges; × = CFD alone



No full Y run admitted

Mean CFD: (80.79, 0.52, -0.02) µm/s
Best tested total X: 80.67 µm/s
Best tested +Z progress: 1.99 µm/s
Target wall distance: 16.50 µm
Target and six neighbors: field/CFD valid = True

Finite candidates at the original F5 300 ms state.
Ranges are not global controllability bounds.
One XZ view is not a 3-D reachability proof.
Free-space RPY has no calibrated wall mobility.